# Assignment 1: Creating a simple Blockchain

## Objective

The objective of this assignment is to implement a simplified blockchain in Go to understand block structure, hash linking, immutability, tamper detection, and chain verification logic. This assignment does NOT include Proof-of-Work.

## Submission Requirements

- Create a Go package named exactly: assignment01bca.
- The package must be uploaded to GitHub.
- Submit a ZIP file named: assignment01bca_YourRollNo.zip.
- The ZIP must contain main.go and the assignment01bca package.
- Deadline: 5th Oct, 11:00 PM. Late or email submissions will receive zero marks.
- You must demonstrate your code in class.

### Required Blockchain Structure

You must implement a Block structure that includes at least the following fields:

```
func NewBlock(transaction string, nonce int, previousHash string)  *block {
        A method to add new block. To keep things simple, you could provide a
        sting of your choice as a transaction (e.g., “bob to alice”). Also, use
        any integer value as a nonce. The CreateHash() method will provide you the
        block Hash value.

func ListBlocks() {
        A method to print all the blocks in a nice format showing block data such
        as transaction, nonce, previous hash, current block hash

func ChangeBlock() {
        function to change block transaction of the given block ref

func VerifyChain() {
        function to verify blockchain in case any changes are made.

func CalculateHash (stringToHash string) {
        function for calculating hash of a block. Hash must be calculated using
SHA-256 and must concatenate: transaction + nonce + previousHash + index +
timestamp.
```

## Custom Requirements

### 1. Roll Number Injection

Each student must include the last 3 digits of their roll number inside the transaction string in at least one block. Failure to include this will result in 10% deduction.

### 2. Personalized Genesis Block

The Genesis block must contain: Transaction = 'Genesis Block - YourRollNo' and Nonce = Sum of digits of your roll number.

### 3. Tamper Demonstration in main.go

- Create at least 3 blocks.
- Print the blockchain.
- Verify the blockchain.
- Tamper with Block 1 using ChangeBlock().
- Run VerifyChain() again to show failure.

## Marking Rubric

| **Component** | **Marks** |
|---|---|
| Correct block structure | 40 |
| Hash calculation (SHA-256) | |
| VerifyChain logic | |
| Tamper detection demonstration | |
| Custom roll-based rules | |
| Code structure & package compliance | 10 |
| Total | 50 |